

**PROGRAM:**

```
#include<stdio.h>

#define SIZE 50

struct Student {

    char name[30];

    int rollno;

    int sub[3];

};

void main() {

    int i, j, max, count, total, n, a[SIZE], ni;

    struct Student st[SIZE];

    printf("Enter how many students: ");

    scanf("%d", &n);

    /* for loop to read the names and roll numbers*/

    for (i = 0; i < n; i++) {

        printf("\nEnter name and roll number for student %d : ", i);

        scanf("%s", &st[i].name);

        scanf("%d", &st[i].rollno);

    }

    /* for loop to read ith student's jth subject*/

    for (i = 0; i < n; i++) {

        for (j = 0; j <= 2; j++) {

            printf("\nEnter marks of student %d for subject %d : ", i, j);

            scanf("%d", &st[i].sub[j]);

        }

    }

    /* (i) for loop to calculate total marks obtained by each student*/

    for (i = 0; i < n; i++) {
```

```

        total = 0;

        for (j = 0; j < 3; j++) {

            total = total + st[i].sub[j];

        }

        printf("\nTotal marks obtained by student %s are %d", st[i].name,total);

        a[i] = total;

    }

    /* (ii) for loop to list out the student's roll numbers who    have secured the highest marks in each subject */
    /* roll number who secured the highest marks */

    for(j = 0; j < 3; j++) {

        max = 0;

        for (i = 0; i < n; i++) {

            if (max < st[i].sub[j]) {

                max = st[i].sub[j];

                ni = i;

            }

        }

        printf("\nStudent %s got maximum marks = %d in Subject : %d",st[ni].name, max, j);

    }

    max = 0;

    for (i = 0; i < n; i++) {

        if (max < a[i]) {

            max = a[i];

            ni = i;

        }

    }

    printf("\n%s obtained the total highest marks.", st[ni].name);

    getch();

```

}

**OUTPUT:**

Enter how many students: 2

Enter name and roll number for student 0 : ARUN

4211101

Enter name and roll number for student 1 : KUMAR

4211102

Enter marks of student 0 for subject 0 : 78

Enter marks of student 0 for subject 1 : 98

Enter marks of student 0 for subject 2 : 67

Enter marks of student 1 for subject 0 : 78

Enter marks of student 1 for subject 1 : 55

Enter marks of student 1 for subject 2 : 57

Total marks obtained by student ARUN are 243

Total marks obtained by student KUMAR are 190

Student ARUN got maximum marks = 78 in Subject : 0

Student ARUN got maximum marks = 98 in Subject : 1

Student ARUN got maximum marks = 67 in Subject : 2

ARUN obtained the total highest marks.